4.3 Messaging

4.3.1 Non-Functional Requirements (NFR's)

315 Messaging is categorized under reliability and extensibility . The key NFR's are:

NFR-ident	Requirement	Reference
NFR-MSG-01	For asynchronous inter-module communication a publish and subscribe messaging pattern shall be used..	
NFR-MSG-02	Publishers and subscribers shall operate independently, for flexibility and scalability	
NFR-MSG-03	The Publish and Subscribe systems shall use topic-based categorization. Subscribers express their interests by subscribing to specific topics. Publishers send messages to designated topics.	
NFR-MSG-04	A broker shall dispatch a published message towards the subscribers of this topic.	
NFR-MSG-05	There shall be a COMMAND based topic, mainly used for controlling the system.	
NFR-MSG-07	There shall be an INCIDENT based topic, mainly used for incident processing, such as errors and warnings	
NFR-MSG-08	A message shall be validated by broker, before publishing, to prevent incorrect messages being further processed.	
NFR-MSG-11	The broker will provide a separate queue for each subscriber: each published message must be copied to every subscriber's queue, providing processing speed isolation. Each subscriber can operate at own speed: a slow or offline subscriber can build up its own backlog without delaying faster subscribers.	
NFR-MSG-12	A module which subscribes for a topic shall receives messages related to that topic.	
NFR-MSG-13	A subscriber to a Topic, can request (as an attribute) for the the last published message to related Topic, to be returned to the subscriber upon subscription.	
NFR-MSG-14	A command publisher can request for a confirmation (as an attribute) that a published command message was processed by a specific subscriber. In case command was not processed correctly, the publisher will be notified.	